

HANGA WORKS

Software Requirements Specification

Skills, Employment & Workforce Intelligence Platform

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Prepared for internal engineering, investor, and enterprise stakeholder review.

1. Introduction

1.1 Purpose

HANGA WORKS is a web-based workforce intelligence and employment platform engineered to unify skills development, verified talent profiling, and structured employment pipelines into a single, data-driven digital ecosystem.

The platform enables individuals to build verifiable competency profiles, progress through structured learning pathways, and connect with employment opportunities through intelligent matching algorithms — replacing the traditional CV-based model with measurable, skills-first hiring.

This document defines the complete functional, non-functional, architectural, and integration requirements governing the design, development, and deployment of the HANGA WORKS platform. It is the authoritative reference for all engineering, product, and delivery decisions.

1.2 Scope

HANGA WORKS operates as a fully web-accessible, SaaS-style platform designed for horizontal scale across education, employment, and enterprise workforce sectors. The platform serves as a centralized ecosystem where learning, talent verification, hiring, and workforce analytics coexist within a unified system.

The platform encompasses the following core functional domains:

- Digital learning and competency development
- Intelligent job and talent marketplace
- Career intelligence and recommendation engine
- Digital certification and credential verification
- Employer recruitment and pipeline management dashboard
- Workforce analytics and labor market intelligence

These domains are architecturally interconnected to create a continuous lifecycle:

Learn → Certify → Match → Employ → Improve

1.3 Strategic Objectives

HANGA WORKS is designed to address a structural gap between education systems and labor markets. The platform's strategic objectives are:

1. Establish skills-based hiring as the dominant recruitment model, replacing fragmented CV-driven processes.
2. Improve access to structured employment pathways for underserved talent populations.
3. Reduce unemployment friction through intelligent, real-time job-to-talent matching.
4. Standardize digital skill verification and credentials across institutions and employers.

5. Support continuous workforce upskilling through personalized learning recommendations.
6. Provide employers with pre-qualified, data-enriched candidate pipelines.
7. Generate actionable workforce intelligence for institutions, employers, and policy stakeholders.

1.4 Value Proposition

HANGA WORKS transforms fragmented education and employment infrastructure into a single intelligent platform where:

Stakeholder	Core Value Delivered
Job Seekers / Learners	Structured skills development, verified credentials, and intelligent job matching
Employers	Access to pre-qualified, skills-verified candidates; reduced time-to-hire
Training Institutions	A digital pipeline to connect graduates directly with employment opportunities
Workforce Planners / Government	Real-time labor market intelligence and skills gap analysis

1.5 Definitions & Key Terms

Term	Definition
User	Any registered individual on the HANGA WORKS platform
Learner	A user actively enrolled in a learning program or course
Employer	An organization using the platform to recruit talent
Skill Profile	A structured, verified digital representation of a user's competencies
Job Match Score	An algorithm-generated compatibility rating between a user and a job listing
Certification	A digitally issued, verifiable credential confirming competency completion
Workforce Intelligence	Aggregated data insights on labor market demand, talent supply, and skill gaps
LMS	Learning Management System — the platform module governing course delivery
ATS	Applicant Tracking System — the module managing employer recruitment workflows
RBAC	Role-Based Access Control — the security model governing user permissions

Term	Definition
SRS	Software Requirements Specification — this document

2. System Overview

2.1 Platform Description

HANGA WORKS is a modular, cloud-ready SaaS platform that replaces fragmented hiring and education workflows with a centralized, data-driven ecosystem. It improves talent visibility, learning outcomes, and recruitment efficiency by connecting all stakeholders — learners, employers, institutions, and administrators — within a single operating environment.

2.2 Architectural Layers

The system is organized into five interconnected architectural layers:

Layer	Function	Key Components
User Layer	Manages all actor identities, roles, and access	Registration, Auth, RBAC, Profiles
Learning Layer	Delivers structured skills development pathways	LMS, Assessments, Certifications
Employment Layer	Manages job discovery, applications, and hiring	Job Marketplace, ATS, Employer Dashboard
Intelligence Layer	Generates recommendations and market insights	Matching Engine, Analytics, Career AI
Administration Layer	Governs platform operations and data integrity	Moderation, Reporting, System Config

2.3 Platform Lifecycle

The system operates as a continuous user journey:

- 8. User registers and completes identity verification
- 9. Skill profile is created via self-assessment and prior certification import
- 10. Interest mapping and career goal setting
- 11. Personalized learning pathway is generated
- 12. User completes courses, assessments, and evaluations
- 13. Digital certification is issued and made verifiable
- 14. Intelligent job matching generates ranked opportunity recommendations
- 15. Employer reviews candidate, initiates hiring workflow
- 16. Post-hire: continuous learning loop is re-engaged for upskilling

2.4 User Roles & Access Model

Role	Description	Core Capabilities
Learner / Job Seeker	Individual using the platform to develop skills and access employment	Learn, certify, apply, track progress
Employer	Organization recruiting talent via the platform	Post jobs, search candidates, manage pipeline
Institution	Educational or training body providing certified content	Create courses, issue credentials, track learners
Mentor	Industry professional offering career guidance	Accept bookings, conduct sessions, review mentees
Administrator	Platform operator managing governance and data	Manage users, content, analytics, configurations

3. Functional Requirements

3.1 User Management & Identity System

Manages secure identity creation, authentication, and role-based access for all platform actors.

3.1.1 Core Features

- Email and phone-based registration with verification flow
- JWT-based session management with refresh token rotation
- Role-Based Access Control (RBAC) with granular permission sets
- Multi-factor authentication (MFA) support
- User profile management: personal info, skills, experience, preferences
- Skill tagging, categorization, and proficiency self-rating
- Account status management: active, suspended, deactivated
- Audit log of user activity and account changes

3.2 Learning Management System (LMS)

Delivers structured, outcome-driven learning experiences with progress tracking and assessment capabilities.

3.2.1 Course Management

- Course creation, publishing, and version control
- Multi-format content: video, documents, interactive modules, SCORM support
- Structured learning path design with prerequisite chaining
- Drafts, review, and approval workflow for institution-published content

3.2.2 Learner Experience

- Personalized learning dashboard with progress metrics
- Course progress tracking and completion milestones
- Quizzes, assessments, and practical evaluations
- Discussion and peer learning forums per course
- Skill-based course recommendations (intelligence-layer driven)
- Offline access support for selected content

3.3 Job Marketplace

Provides a structured, searchable environment for job discovery, application, and tracking.

3.3.1 Job Discovery

- Job posting with structured data: title, description, required skills, salary, type
- Advanced search and filter: skill, location, job type, salary range, experience level
- Remote, hybrid, on-site, internship, and freelance listings
- Employer-branded job pages
- Recommended jobs powered by the matching engine

3.3.2 Application Management

- One-click application using Skill Profile
- Application status tracking: applied, reviewing, shortlisted, rejected, hired
- Saved jobs and alert subscriptions
- Application history and timeline view

3.4 Career Intelligence Engine

Provides users with data-driven career direction and personalized growth recommendations.

- Skill gap analysis: compares current profile to target role requirements
- Career pathway mapping: visualizes progression routes to target roles
- Industry trend insights: highlights high-demand skills and sectors
- Recommended next steps: courses, certifications, and experiences
- Salary benchmarking by role, location, and experience level
- Job role compatibility scoring based on current profile

3.5 Certification & Verification System

Ensures trust, portability, and verifiability of all platform-issued credentials.

- Automated digital certificate generation upon course or program completion
- Unique certificate IDs with tamper-evident verification links
- Skill badge system with shareable public profile integration
- Employer-facing certificate validation portal
- Institution-issued credentials with institutional branding
- Certificate expiry and renewal tracking for time-sensitive credentials

3.6 Employer Recruitment Dashboard

Provides organizations with end-to-end hiring workflow tools integrated with the platform's talent pool.

- Job posting creation, scheduling, and management
- Skills-based candidate search and discovery
- AI-ranked candidate shortlists based on job requirements
- Applicant Tracking System (ATS): manage candidates across pipeline stages
- Talent pipeline management: save, tag, and revisit candidate profiles
- Employer analytics: application volume, time-to-shortlist, diversity metrics
- Team collaboration: multiple recruiter seats with role-scoped access

3.7 Mentorship System

Connects learners with industry professionals for structured career guidance.

- Mentor profile creation: expertise, availability, bio, and experience
- Mentor discovery with search and category filters
- Session booking and calendar scheduling integration
- In-platform messaging and video call link sharing
- Session notes, goals, and progress tracking per mentee
- Mentor ratings and review system

3.8 Notification & Engagement System

Drives real-time engagement across all user types with contextual, targeted alerts.

Trigger Event	Delivery Channel	Target User
New matching job posted	In-app + Email	Learner / Job Seeker
Application status updated	In-app + Email + SMS	Learner / Job Seeker
Course completion	In-app + Email	Learner
New applicant received	In-app + Email	Employer
Certification issued	In-app + Email	Learner
Mentor session reminder	In-app + Email + SMS	Learner + Mentor
Platform announcements	In-app	All Users

3.9 Analytics & Reporting System

Provides platform-wide intelligence for operational, strategic, and compliance reporting.

- User engagement metrics: DAU, MAU, retention, session depth
- Job market trend analysis: demand by skill, sector, and location
- Skill demand heatmaps: identify emerging and declining skill categories
- Course performance metrics: completion rates, assessment scores, drop-off points
- Employer activity reports: post volume, application rates, conversion
- Platform-wide performance dashboards for administrators
- Exportable reports in CSV, PDF, and Excel formats

4. Intelligence Layer

The Intelligence Layer is the analytical core of HANGA WORKS. It processes behavioral, profile, and market data to generate personalized recommendations, matching scores, and workforce insights across all platform modules.

4.1 Job Matching Engine

Generates ranked job recommendations for each learner based on a multi-factor scoring model.

Signal	Weight	Description
Skill alignment	High	Overlap between user skills and job required skills
Experience level	High	Years of experience vs. role requirement
Certification match	Medium	Presence of required or preferred certifications
Location preference	Medium	User location preference vs. job location/remote policy
Engagement behaviour	Low	Jobs saved, viewed, and applied to previously
Career goal alignment	Low	Fit with user's stated career trajectory

4.2 Learning Recommendation Engine

Suggests the most relevant courses and learning paths to each user based on their goals and gaps.

- Skill gap analysis against target roles or industry benchmarks
- Behavioral signals: courses started, completed, dropped, searched
- Peer benchmarking: what learners with similar profiles are completing
- Employer demand signals: courses linked to in-demand job skills

4.3 Workforce Insight Engine

Aggregates platform-wide data to produce labor market intelligence reports for employers, institutions, and administrators.

- Real-time skill demand tracking across active job listings
- Talent supply analysis by skill category, location, and qualification
- Emerging skill trend identification and alert system
- Regional workforce gap reporting

4.4 Predictive Career Modelling

Forecasts future career trajectories and role viability based on market trend data.

- Role viability scoring: likelihood of employment given current market demand
- Skill deprecation alerts: flags skills trending out of employer demand
- Suggested pivot pathways: alternative roles compatible with existing skill sets

5. Non-Functional Requirements

Category	Requirement	Target / Standard
Performance	Page load time	< 2 seconds (P95) on standard broadband
Performance	API response time	< 500ms for standard requests
Performance	Real-time updates	WebSocket latency < 1 second
Security	Data transmission	TLS 1.2+ enforced on all endpoints
Security	Authentication	JWT with refresh rotation; MFA support
Security	Access control	RBAC with least-privilege enforcement
Security	Data privacy	GDPR-aligned data handling; right to erasure
Scalability	Architecture	Horizontally scalable microservice-ready design
Scalability	Cloud infrastructure	Cloud-native deployment (containerized)
Scalability	Concurrent users	Designed for 10,000+ concurrent sessions
Reliability	System availability	99.5% uptime SLA target
Reliability	Data backups	Automated daily backups with point-in-time recovery
Reliability	Fault tolerance	Graceful degradation under component failure
Usability	UI/UX	Clean, accessible, responsive design (WCAG 2.1 AA)
Usability	Onboarding	User activated within < 5 minutes from registration
Usability	Mobile support	Fully responsive for all screen sizes
Maintainability	Codebase	Modular architecture with documented APIs
Maintainability	Monitoring	Real-time error tracking and alerting

6. System Architecture

6.1 Technology Stack

Layer	Technology	Rationale
Frontend	React.js (TypeScript)	Component-based, scalable UI with strong ecosystem support
Backend API	Node.js — NestJS framework	Modular, enterprise-grade structure with built-in DI and testing
Primary Database	PostgreSQL	Relational integrity for structured user, job, and certification data
Cache Layer	Redis	Session management, rate limiting, and real-time data caching
Real-time Layer	WebSockets (Socket.io)	Live notifications, application status updates, chat
File Storage	Cloud Object Storage (S3-compatible)	Certificates, course media, profile assets
Search Engine	Elasticsearch (planned)	Fast full-text job and candidate search at scale
Containerization	Docker + Kubernetes (planned)	Portable, scalable deployment environment

6.2 API Design

- RESTful API design adhering to OpenAPI 3.0 specification
- All endpoints return structured JSON responses with standardized error codes
- API versioning via URI path (e.g. /api/v1/)
- Rate limiting enforced per user and per endpoint
- WebSocket channels for real-time notifications and status updates

6.3 Security Architecture

- HTTPS enforced platform-wide; HTTP requests redirected
- JWT access tokens (15-minute expiry) with refresh token rotation
- Secrets managed via environment variables and secrets manager — never hardcoded
- SQL injection, XSS, and CSRF protections applied at framework level
- Input validation and sanitization on all API endpoints
- Audit logging of all sensitive operations

7. External Integrations

Integration	Purpose	Phase
Email Service (SendGrid / SES)	Transactional emails, alerts, notifications	Phase 1
SMS Gateway (Africa's Talking / Twilio)	SMS alerts for applications and sessions	Phase 1
Payment Gateway (Stripe / Flutterwave)	Course purchases and premium subscriptions	Phase 2
Calendar APIs (Google / Outlook)	Mentor session scheduling integration	Phase 2
Job Board APIs (LinkedIn, etc.)	Inbound job feed expansion	Phase 3
HR Systems (SAP SuccessFactors, etc.)	Enterprise ATS integration	Phase 3
National Skills Registry	Credential portability and official recognition	Phase 3
LMS Standards (SCORM / xAPI)	Third-party course content import	Phase 2

8. Implementation Roadmap

Phase	Scope	Target Milestone
Phase 1 — Core Platform	User management, LMS, basic job marketplace, certification, notifications	Month 1–4
Phase 2 — Intelligence Layer	Matching engine, career intelligence, employer dashboard, mentorship	Month 5–8
Phase 3 — Analytics & Scale	Workforce analytics, advanced AI matching, payment integration, API expansions	Month 9–12
Phase 4 — Enterprise & Global	HR integrations, national registry, cross-border expansion, white-labelling	Month 13+

9. Success Metrics & KPIs

Metric	Definition	Target (Year 1)
Monthly Active Users (MAU)	Unique users active in a 30-day window	10,000+
Job Placement Rate	Learners who secured employment via the platform	25%+
Employer Adoption Rate	Active employers posting jobs in a given month	100+
Course Completion Rate	Enrolled users who complete a full course	> 60%
Matching Accuracy	User-rated relevance of recommended jobs	> 75% positive
Time-to-Hire Reduction	Vs. industry average for employer users	30% reduction
Platform Retention Rate	Users active 90 days after registration	> 40%
Certification Issue Rate	Learners who earn at least one credential	> 50%

10. Future Enhancements

- Advanced generative AI for personalized career coaching and interview preparation
- Integrated video interview and async screening module
- Global freelance and gig marketplace with escrow payment support
- National and regional skills registry integration for formal credential recognition
- Automated hiring workflows: AI-driven screening, scheduling, and offer management
- White-label platform offering for governments and large enterprises
- Cross-border job matching with work permit guidance and compliance tools
- Employer diversity and inclusion analytics module
- Mobile application (iOS and Android) for learner and employer access

11. Conclusion

HANGA WORKS is engineered to become a critical piece of digital infrastructure for workforce development and employment efficiency. By replacing fragmented, credential-blind hiring processes with a unified, intelligence-driven platform, HANGA WORKS creates measurable value for every stakeholder in the employment ecosystem.

The platform is designed with scalability, security, and institutional trust at its core — making it ready for enterprise adoption, government partnership, and international expansion from day one.

This SRS defines the complete technical and functional blueprint for the HANGA WORKS platform and serves as the governing document for all product, engineering, and delivery decisions throughout its development lifecycle.

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